

BLUESTOCK MUTUAL FUND ANALYTICS PROJECT



BLUESTOCK INTERNSHIP PROJECT

SUBMITTED BY:

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TOOLS USED:

PYTHON | PANDAS | SQLITE | POWER BI | JUPYTER NOTEBOOK

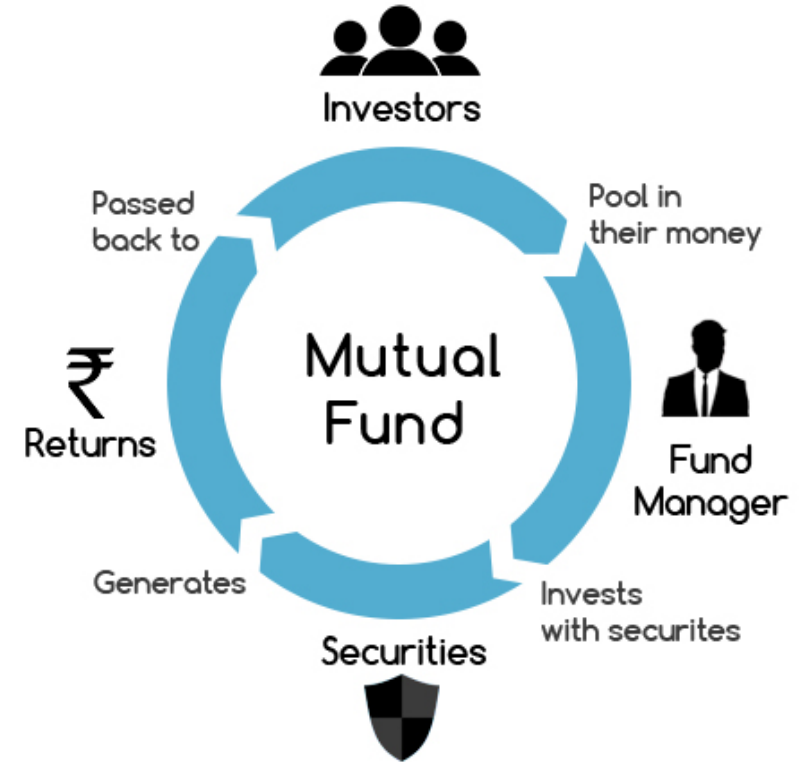
PROBLEM STATEMENT

The Indian Mutual Fund industry has experienced rapid growth in assets under management (AUM), SIP inflows, and investor participation.

However, investors face challenges in:

- Comparing fund performance
- Understanding risk
- Tracking market trends
- Selecting suitable funds

The objective of this project is to build a complete analytics solution that transforms mutual fund data into actionable insights.



PROJECT OBJECTIVES

- Build an ETL pipeline for mutual fund datasets
- Design a structured SQLite database
- Perform Exploratory Data Analysis (EDA)
- Calculate performance metrics such as:
 - CAGR
 - Sharpe Ratio
 - Sortino Ratio
 - Alpha & Beta
- Develop advanced risk analytics:
 - VaR
 - CVaR
 - Rolling Sharpe
- Build an interactive Power BI dashboard
- Generate fund recommendations



DATA SOURCES



Datasets:

- Fund Master
 - NAV History
 - AUM by Fund House
 - Monthly SIP Inflows
 - Category Inflows
 - Industry Folio Count
 - Scheme Performance
 - Investor Transactions
 - Portfolio Holdings
 - Benchmark Indices
- Total Datasets Used: 10
 - Total Mutual Fund Schemes: 40

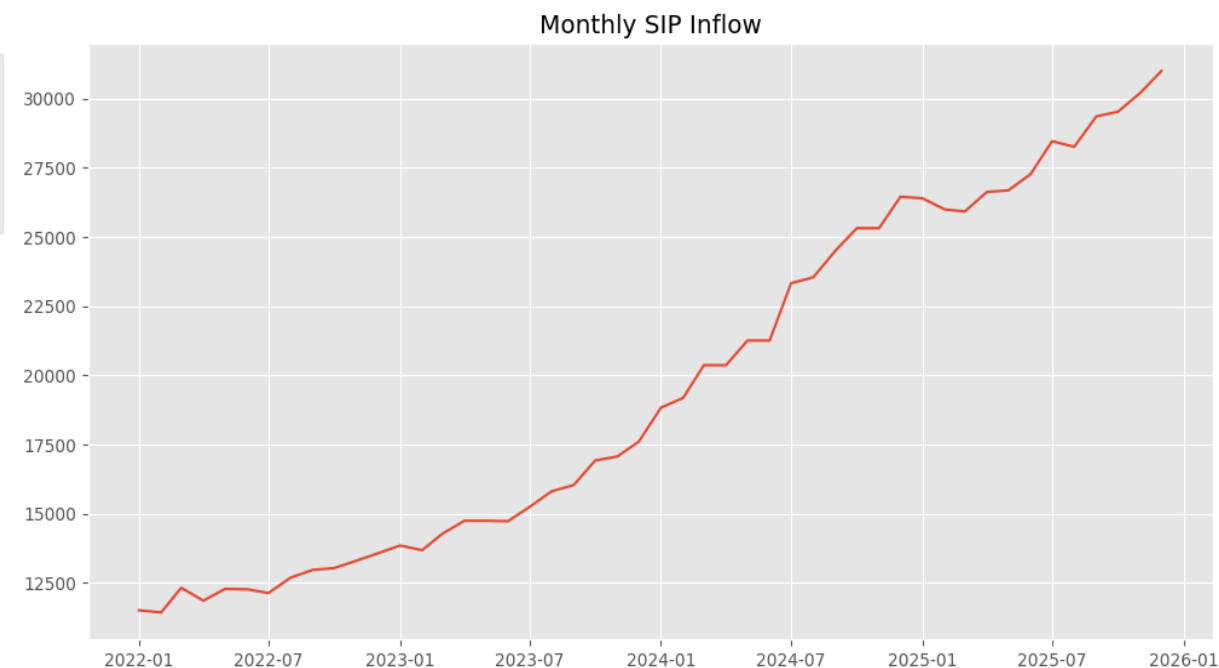
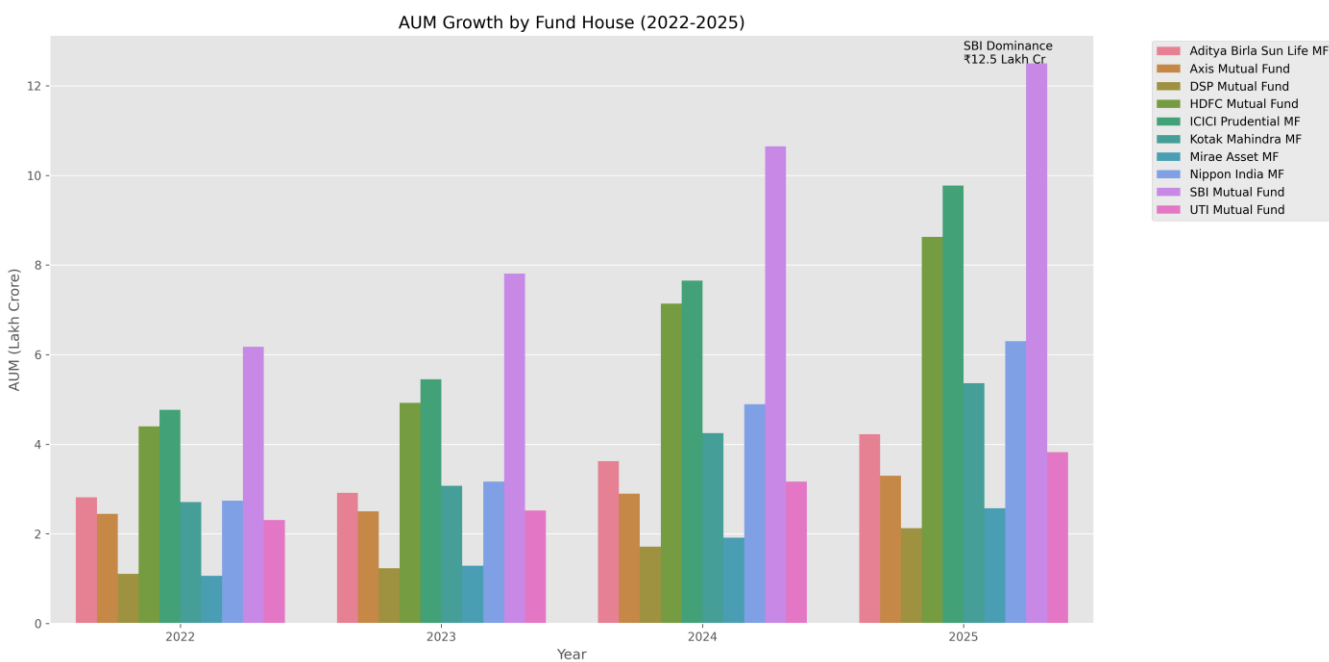
PROJECT ARCHITECTURE



EDA HIGHLIGHTS



- SIP inflows increased steadily between 2022 and 2025
- AUM growth shows strong expansion of the mutual fund industry
- Equity-oriented schemes dominate investor interest
- Industry folio counts nearly doubled during the study period

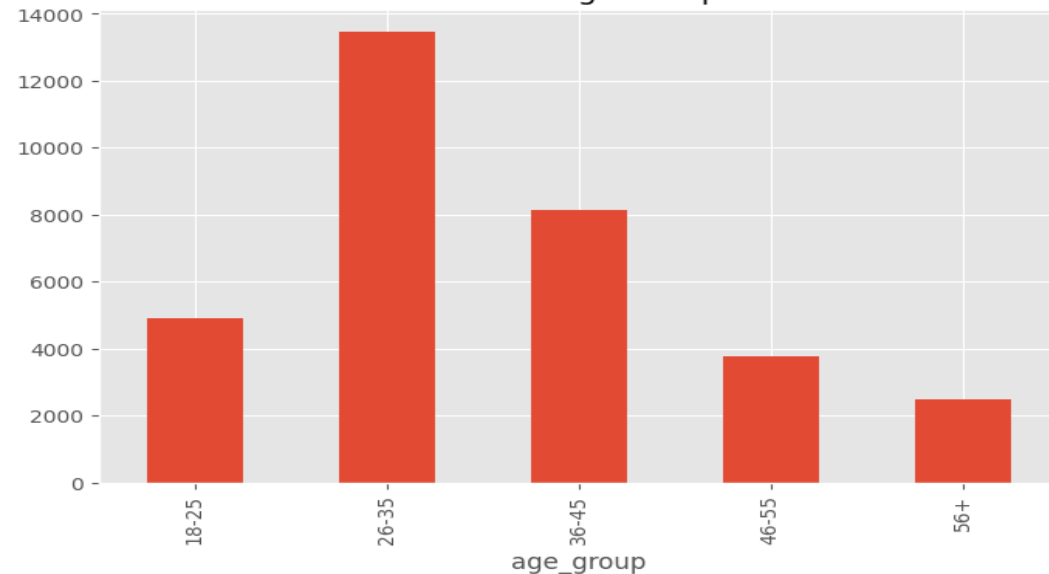


INVESTOR ANALYTICS

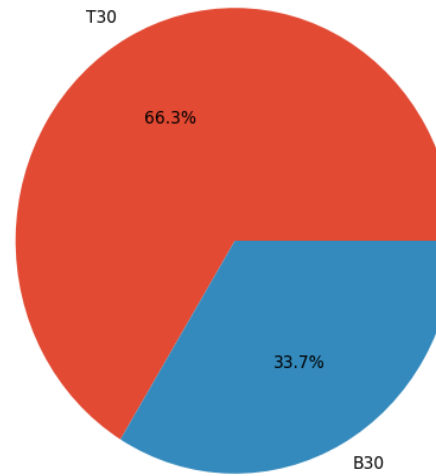


- Investors are concentrated in major states
- T30 cities contribute significant investment volume
- SIP is the most preferred investment mode
- Younger investors show increasing participation

Investor Age Groups



T30 vs B30 Investors



Transaction Type Distribution

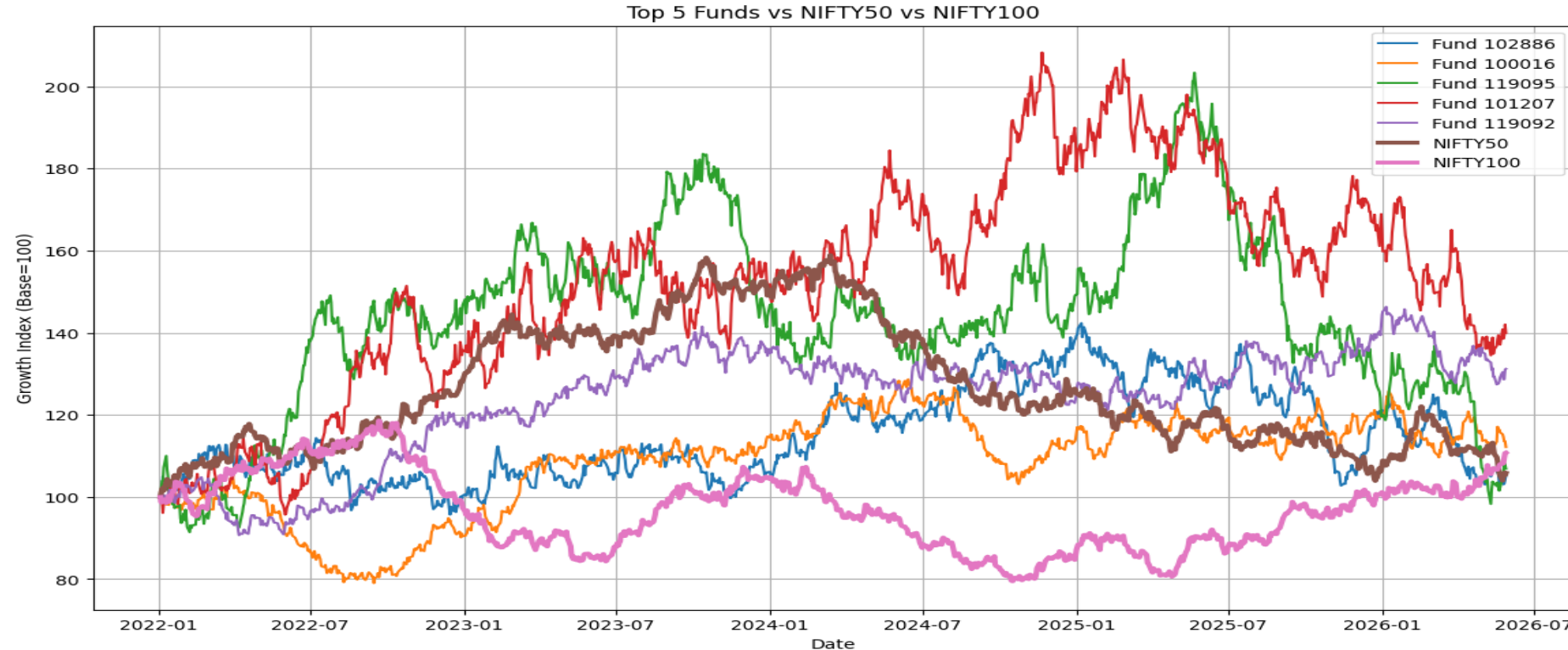


FUND PERFORMANCE ANALYTICS



Metrics Calculated:

- CAGR (1Y, 3Y, 5Y)
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Tracking Error
- Composite Fund Score



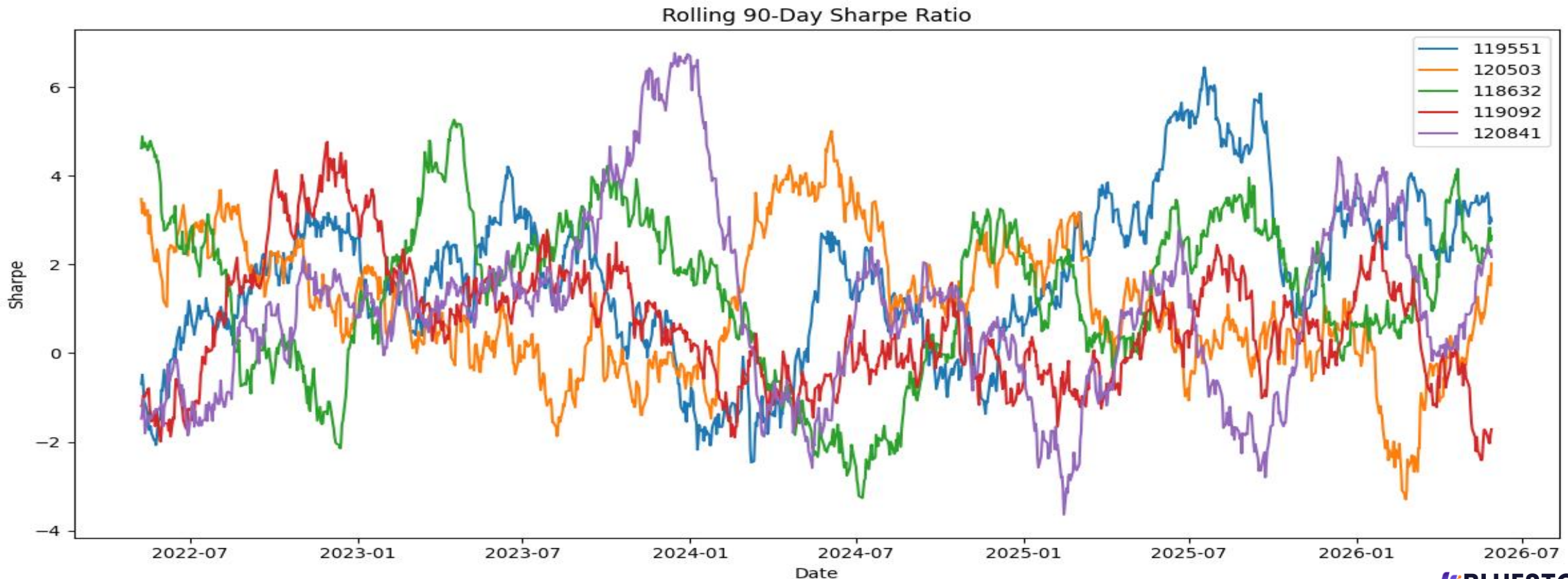
- **Fund 101207 delivered the strongest overall performance**, peaking above **200 (Base = 100)** during 2024–2025 and finishing well above most other funds, indicating superior long-term growth.
- **Fund 119095 showed high growth but also high volatility**, reaching levels above **180–200** before experiencing significant drawdowns, suggesting higher return potential with greater risk.
- **Most top funds outperformed NIFTY50 and NIFTY100 over the analysis period**, particularly Funds **101207, 119095, and 119092**, demonstrating stronger wealth creation than the benchmark indices.

ADVANCED RISK ANALYTICS



Risk Metrics Implemented:

- Historical VaR , Rolling 90-Day Sharpe Ratio , HHI Portfolio Concentration Index
- Conditional VaR (CVaR) , SIP Continuity Analysis , Investor Cohort Analysis



DASHBOARD OVERVIEW



1. Industry Overview
2. Investor Analytics
3. Fund Performance
4. Sip & Market Trends

Features:

- Interactive slicers
- KPI cards
- Fund drill-through
- Trend analysis
- Investor analytics

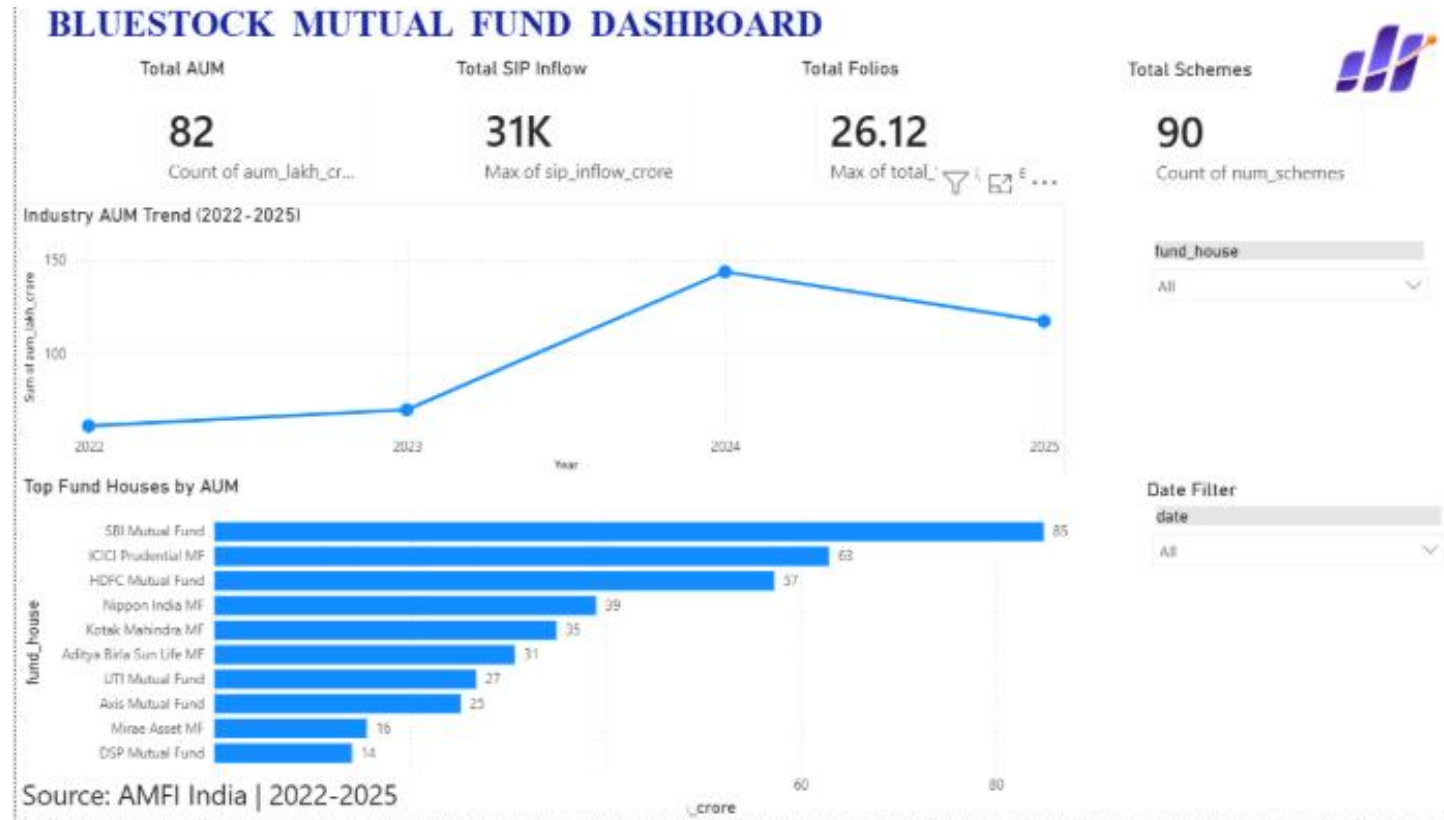


Power BI

POWER BI DASHBOARD [INDUSTRY OVERVIEW]



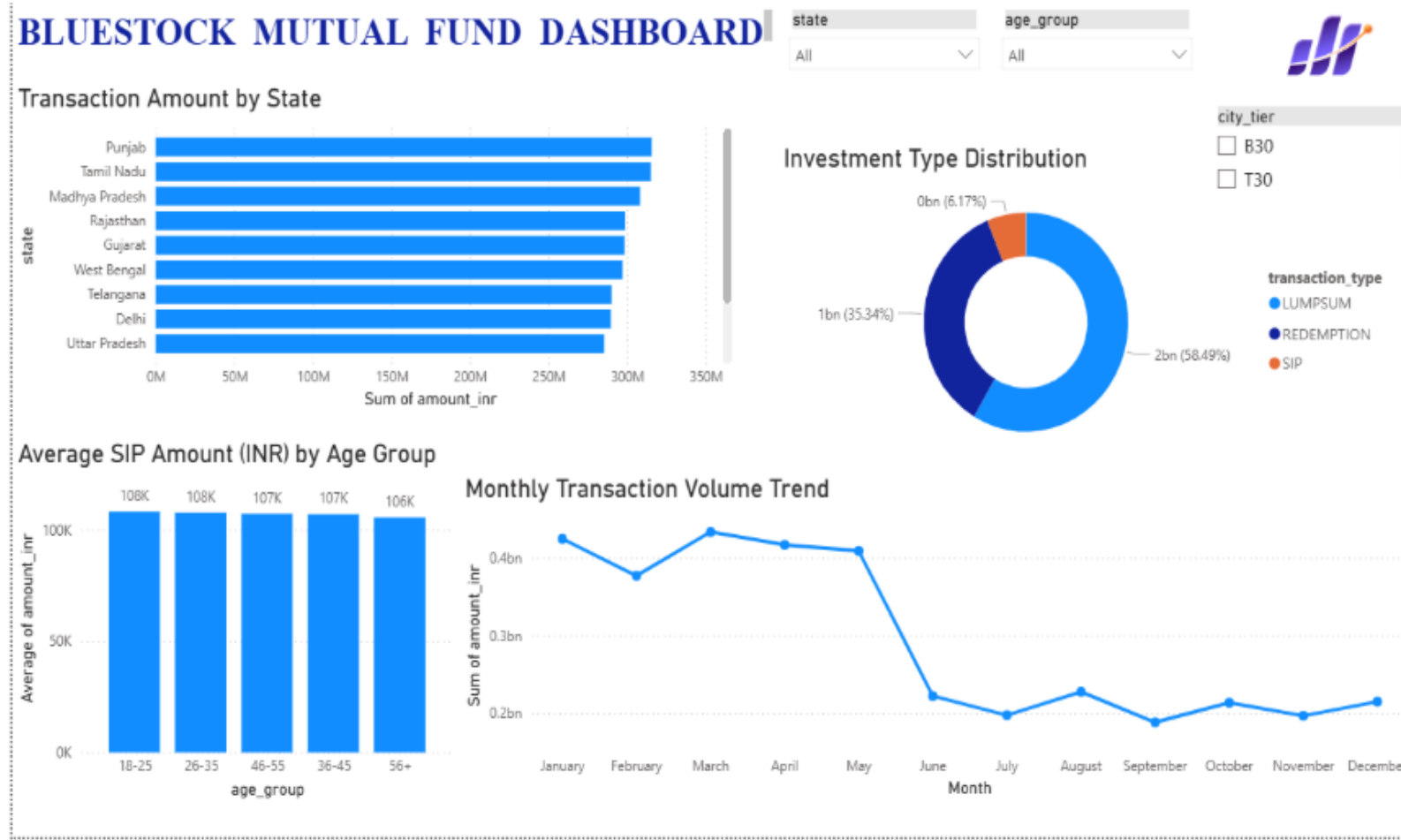
- **Industry AUM grew strongly between 2022 and 2024**, indicating rising investor participation and expanding opportunities for asset management companies.
- **Market share is concentrated among a few large fund houses**, with SBI Mutual Fund, ICICI Prudential, and HDFC Mutual Fund leading in AUM, highlighting a highly competitive but concentrated industry structure.
- **Strong SIP inflows and a large scheme base** suggest sustained retail investor interest, supporting long-term growth potential for the mutual fund industry.



POWER BI DASHBOARD [INVESTOR ANALYTICS]



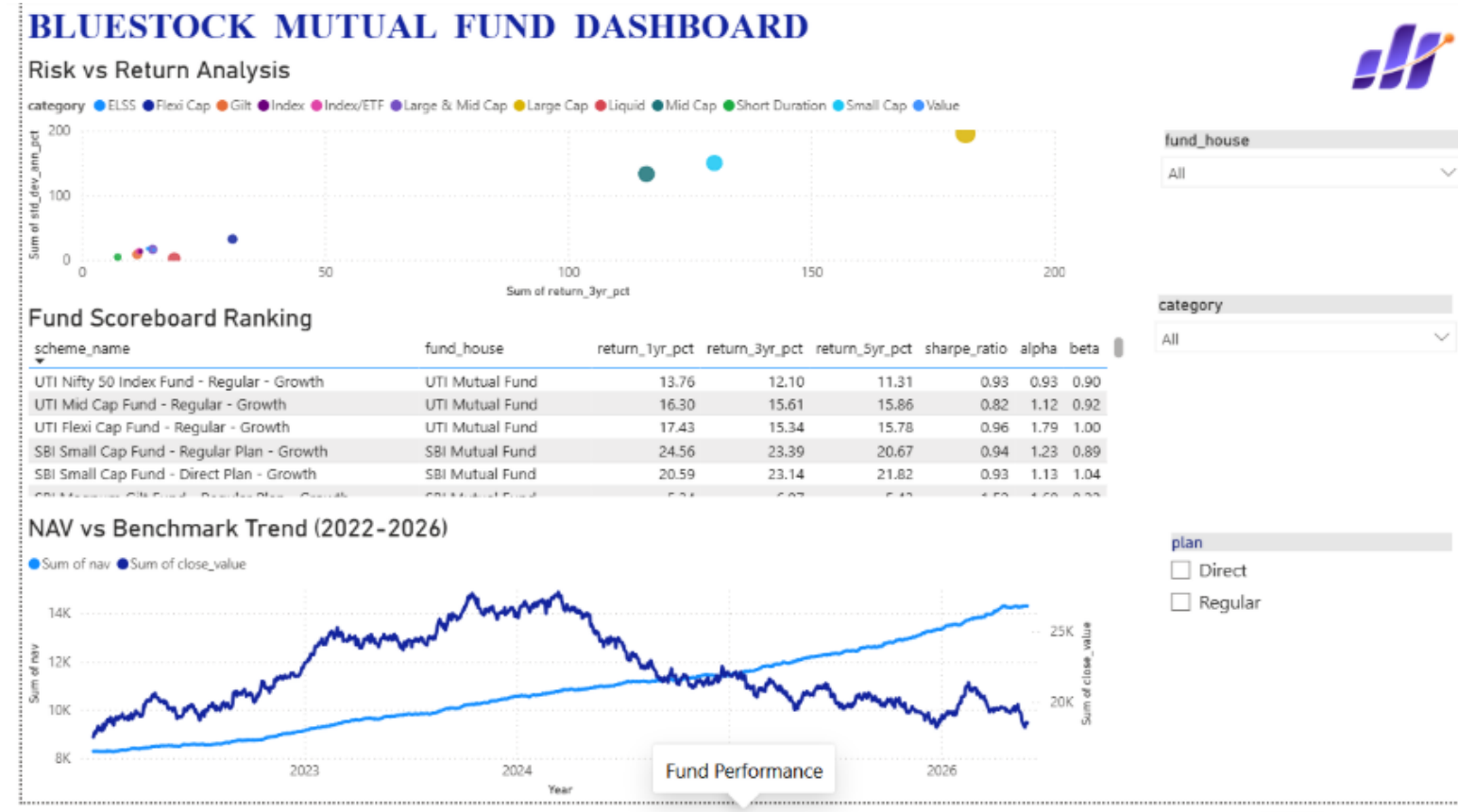
- **Investment participation is geographically diversified**, with states such as Punjab, Tamil Nadu, and Madhya Pradesh contributing the highest transaction volumes, indicating broad retail investor engagement across regions.
- **Lumpsum investments dominate transaction value (~58%)**, significantly exceeding SIP and redemption volumes, suggesting investors are allocating larger one-time investments during the analysis period.
- **Average SIP amounts are consistent across age groups (~₹106K–₹108K)**, showing that mutual fund adoption and investment capacity are relatively balanced among different generations of investors.



POWER BI DASHBOARD [FUND PERFORMANCE]



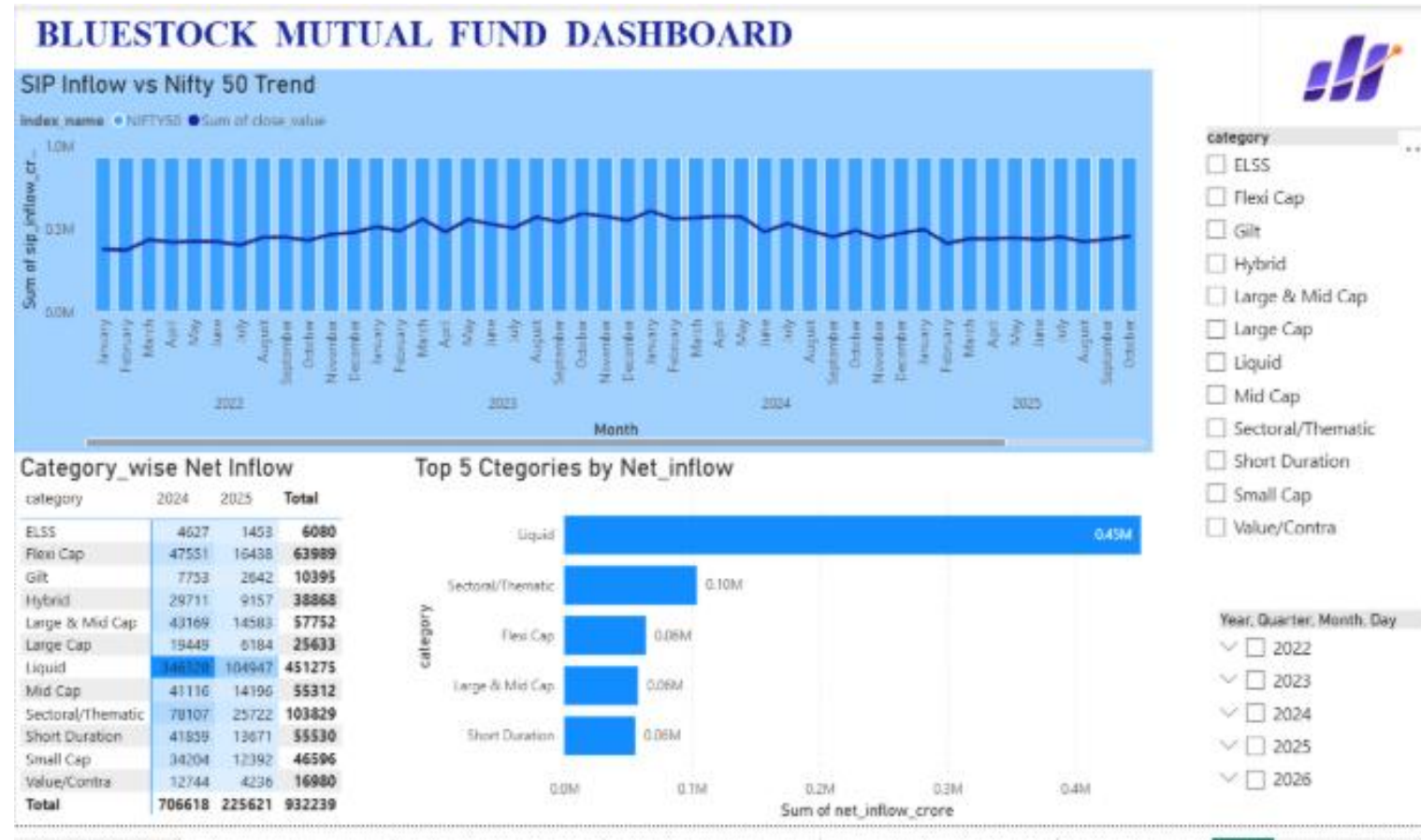
- **Higher-risk fund categories generally delivered higher returns**, as shown in the Risk vs Return analysis, indicating a positive risk–reward relationship for investors seeking long-term growth.
- **Several UTI and SBI schemes rank among the top-performing funds**, demonstrating strong consistency across return metrics, Sharpe Ratio, Alpha, and Beta scores.
- **Benchmark indices showed steady long-term growth from 2022–2026**, while individual fund NAVs experienced greater fluctuations, highlighting the importance of diversification and active fund selection.



POWER BI DASHBOARD [SIP & MARKET TRENDS]



- **SIP inflows remained relatively stable from 2022–2025**, indicating consistent retail investor participation despite fluctuations in the NIFTY 50 index.
- **Liquid Funds attracted the highest net inflows (≈0.45M crore)**, significantly outperforming other categories, suggesting investors prioritized liquidity and lower-risk investment options.
- **Sectoral/Thematic and Flexi Cap categories emerged as strong growth segments**, reflecting increasing investor interest in diversified and theme-driven investment strategies.



KEY FINDINGS & RECOMMENDATIONS



FINDINGS:

- Several funds achieved superior Sharpe Ratios indicating better risk-adjusted returns.
 - SIP inflows showed consistent growth over the analysis period.
 - High HHI funds demonstrate greater portfolio concentration risk.
 - Investor cohorts entering in recent years contributed larger investment amounts.
 - Funds with high VaR values carry greater downside risk.
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RECOMMENDATIONS:

- Prefer funds with strong Sharpe Ratios.
- Monitor concentration risk using HHI.
- Improve SIP continuity among at-risk investors.
- Use risk-grade based fund recommendations.
- Track rolling Sharpe ratios for performance monitoring.



THANK YOU
